

Docker-03

1. Create a image from running container.

- Connect with EC2

```
Arunasai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ ssh -i "aws.pem" ubuntu@ec2-3-90-180-17.compute-1.amazonaws.com
The authenticity of host 'ec2-3-90-180-17.compute-1.amazonaws.com (3.90.180.17)'
can't be established.
ED25519 key fingerprint is SHA256:EucMPpJ/1hn1NmVKHXzLMv5Chb2XSRYR2SZHCov3Kwo.
This host key is known by the following other names/addresses:
  ~/.ssh/known_hosts:137: ec2-3-87-91-37.compute-1.amazonaws.com
  ~/.ssh/known_hosts:140: ec2-184-72-142-178.compute-1.amazonaws.com
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-3-90-180-17.compute-1.amazonaws.com' (ED25519) t
o the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1004-aws x86_64)
```

- Checked container and created image for running container

```
ubuntu@ip-172-31-25-147:~$ docker container ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS          NAMES
ubuntu@ip-172-31-25-147:~$ docker commit custom-nginx-container custom-nginx-back
kup:v1
sha256:22764ef7122821e4ed4f5bd9b095c1fe16a267bd22a8c16fa1d9888c3f73a0f4
```

- Custom – nginx image created

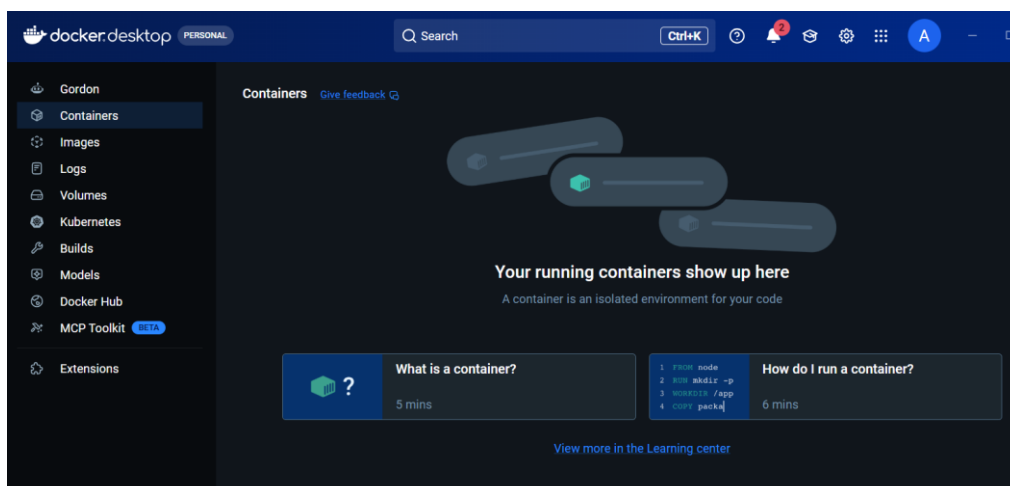
```
ubuntu@ip-172-31-25-147:~$ docker images
```

IMAGE	ID	DISK USAGE	Info + U	In Use
928430096503.dkr.ecr.us-east-1.amazonaws.com/aws.configure:latest				
amazonlinux:latest	371e84608aa7	584MB	195MB	U
arunsail7/techie19:v1	ad882807c6c8	227MB	59.3MB	
custom-nginx-backup:v1	925917d788e0	584MB	195MB	
custom-nginx:v1	22764ef71228	584MB	195MB	
hello-world:latest	371e84608aa7	584MB	195MB	U
jenkins/jenkins:latest	0e760fdfbcb48	25.9kB	9.49kB	U
nginx:latest	d566c401e890	801MB	290MB	U
sonarqube:latest	5aca99593157	241MB	66MB	U
tomcat:9	35bedac3f40c	2.52GB	1.08GB	
tomcat:latest	5f08b8b793ed	592MB	158MB	U
ubuntu:latest	247650787b97	589MB	158MB	
ubuntu:latest	f3d28607ddd7	160MB	45.3MB	

```
ubuntu@ip-172-31-25-147:~$
```

2. Copy image from local machine to docker server and load the image.

- For this first Install Docker Desktop in our local machine



- `docker --version`

```
C:\Users\LENOVO>docker --version
Docker version 29.5.2, build 79eb04c
```

- Docker pull tomcat

```
C:\Users\LENOVO>docker run -itd -p 81:80 tomcat:latest
Unable to find image 'tomcat:latest' locally
latest: Pulling from library/tomcat
d264b91a55d8: Pull complete
4f4fb700ef54: Pull complete
8081bedbbfee: Pull complete
28e0126b19f1: Pull complete
cb259a83ac3d: Pull complete
72c6a52a9f2d: Pull complete
60881f5d8c56: Pull complete
874f21211405: Download complete
a1d8dadfa09f: Download complete
Digest: sha256:f8d2287df1bb2bf9b75a6acfc6d25a75b3a9044429906ff04e01533cfbafa028
Status: Downloaded newer image for tomcat:latest
f0f23ffe4c32b9ef92bbef6fb535e36e0dfdcc2ac4c2feabef8abe2c4e19f073
```

- As we cannot copy this image to our server, we will be creating a tar file using command
Docker save -o tomcat.tar tomcat :latest

```
PS C:\Users\LENOVO\Downloads> docker save -o tomcat.tar tomcat:latest
PS C:\Users\LENOVO\Downloads> ls

Directory: C:\Users\LENOVO\Downloads

Mode                LastWriteTime         Length Name
----                -
-a----             11-05-2026   23:31         519032 01_AWS_Lambda.pptx
-a----             11-05-2026   23:35         327019 02_AWS_API_Gateway.pptx
-a----             11-05-2026   23:38         270981 03_AWS_KMS.pptx
-a----             11-05-2026   23:42         367859 04_AWS_SQS.pptx
```

- As we can see from above tar file is created now let's move the tar file to our server using command

scp -i aws.pem tomcat.tar ubuntu@ec2-54-226-179-195.compute-1.amazonaws.com:/home/ubuntu

```
PS C:\Users\LENOVO\Downloads> scp -i "aws.pem" tomcat.tar ubuntu@ec2-54-226-179-195.compute-1.amazonaws.com:
tomcat.tar                                                    100% 151MB 2.4MB/s
PS C:\Users\LENOVO\Downloads>
```

- Connect to EC2
ssh -i "aws.pem" ubuntu@ec2-54-226-179-195.compute-1.amazonaws.com

```
PS C:\Users\LENOVO\Downloads> ssh -i "aws.pem" ubuntu@ec2-54-226-179-195.compute-1.amazonaws.com
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)
```

```
* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro
```

System information as of Wed Jun 3 10:46:39 UTC 2026

```
System load:  0.0      Temperature:    -273.1 C
Usage of /:   86.7% of 18.25GB    Processes:      161
Memory usage: 13%      Users logged in: 0
Swap usage:   0%        IPv4 address for ens5: 172.31.25.147
```

➤ Verify TAR File

```
Last login: Wed Jun 3 10:18:54 2026 from 18.206.107.28
ubuntu@ip-172-31-25-147:~$ ls
Python-app  alpine-tomcat  custom-nginx.tar  docker-nginx  index.html  multi-stage-example  test.tar  tomcat.tar
ubuntu@ip-172-31-25-147:~$ docker load -i tomcat.tar
Loaded image: tomcat:latest
```

➤ Verify Image

```
ubuntu@ip-172-31-25-147:~$ docker images
```

IMAGE	ID	DISK	USAGE	C
928430096503.dkr.ecr.us-east-1.amazonaws.com/aws.configure:latest	371e84608aa7		584MB	
928430096503.dkr.ecr.us-east-1.amazonaws.com/docker-scan-demo:latest	371e84608aa7		584MB	
alpine-tomcat:v1	f61ad2630e2b		494MB	
alpine:latest	5b10f432ef3d	13.1MB		
amazonlinux:latest	ad882807c6c8		227MB	
arunsail7/custom-nginx:v1	9943b2ca5760		238MB	
arunsail7/techie19:v1	925917d788e0		584MB	
custom-nginx-backup:v1	22764ef71228		584MB	
custom-nginx:v1	371e84608aa7		584MB	
eclipse-temurin:17-jre	0d79988c6879		430MB	
hello-world:latest	0e760dfd4bc48	25.9kB		
jenkins/jenkins:latest	d566c401e890		801MB	
jenkins/jenkins:lts	01c992ffef29		799MB	
maven:3.9.6-eclipse-temurin-17	29a1658b1f30		728MB	
multi-stage-app:v1	257ec9a46506		462MB	
nginx:latest	5aca99593157		241MB	
python-app:v1	cdd3b51b9ef7	1.34GB		
python:3.6	f8652afaf88c	1.32GB		
single-stage-app:v1	80b75d803214	871MB		
sonarqube:latest	35bedac3f40c	2.52GB		
test-nginx:latest	1ef27cf52923	238MB		
tomcat:9	5f08b8b793ed	592MB		

➤ Run container

```
ubuntu:latest
ubuntu@ip-172-31-25-147:~$ docker run -itd -p 8086:8080 tomcat:latest
fa883ac62cd6986e636718118942b4fa1881acbf5c516e28012df9d36202f275
ubuntu@ip-172-31-25-147:~$
```

➤ Docker Container has been created and working on the given port

← → ↻ ⚠ Not secure 54.226.179.195:8086

HTTP Status 404 – Not Found

Type Status Report

Description The origin server did not find a current representation for the target resource or is not willing to disclose that one exists.

Apache Tomcat/11.0.22

3. Create Docker image using alpine and customize with tomcat.

- Create Project Directory and Docker file

```
Last login: Tue Jun 2 11:45:37 2020 from 18.208.107.27
ubuntu@ip-172-31-25-147:~$ mkdir alpine-tomcat
mkdir: alpine-tomcat: File exists
ubuntu@ip-172-31-25-147:~$ cd alpine-tomcat
ubuntu@ip-172-31-25-147:~/alpine-tomcat$ vi Dockerfile
```

- In Docker file added this data

```
FROM alpine:latest
LABEL maintainer="Arunsaai"

RUN apk update && \
    apk add openjdk11 wget tar

RUN wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.96/bin/apache-tomcat-9.0.96.tar.gz && \
    tar -xzf apache-tomcat-9.0.96.tar.gz && \
    mv apache-tomcat-9.0.96 /tomcat

EXPOSE 8080

CMD ["/tomcat/bin/catalina.sh", "run"]
```

- Save and Verify Docker file

```
ubuntu@ip-172-31-25-147:~/alpine-tomcat$ vi Dockerfile
ubuntu@ip-172-31-25-147:~/alpine-tomcat$ ls
Dockerfile
ubuntu@ip-172-31-25-147:~/alpine-tomcat$
```

- Build Docker Image

```
ubuntu@ip-172-31-25-147:~/alpine-tomcat$ docker build -t alpine-tomcat:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  2.048kB
Step 1/6 : FROM alpine:latest
latest: Pulling from library/alpine
6a0ac1617861: Pulling fs layer
6a0ac1617861: Download complete
```

- Successfully Build

```
---> 8d6a4603d5e2
Step 6/6 : CMD ["/tomcat/bin/catalina.sh", "run"]
---> Running in 3db2180c6262
---> Removed intermediate container 3db2180c6262
---> f61ad2630e2b
Successfully built f61ad2630e2b
Successfully tagged alpine-tomcat:v1
ubuntu@ip-172-31-25-147:~/alpine-tomcat$
```


➤ Verify docker images

```
ubuntu@ip-172-31-25-147:~/alpine-tomcat$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
928430096503.dkr.ecr.us-east-1.amazonaws.com/aws.configure:latest	371e84608aa7	584MB	195MB	U
alpine-tomcat:v1	f61ad2630e2b	494MB	178MB	
alpine:latest	5b10f432ef3d	13.1MB	3.95MB	
amazonlinux:latest	ad882807c6c8	227MB	59.3MB	
arunsail7/techie19:v1	925917d788e0	584MB	195MB	
custom-nginx-backup:v1	22764ef71228	584MB	195MB	
custom-nginx:v1	371e84608aa7	584MB	195MB	U
hello-world:latest	0e760fd7bc48	25.9kB	9.49kB	U

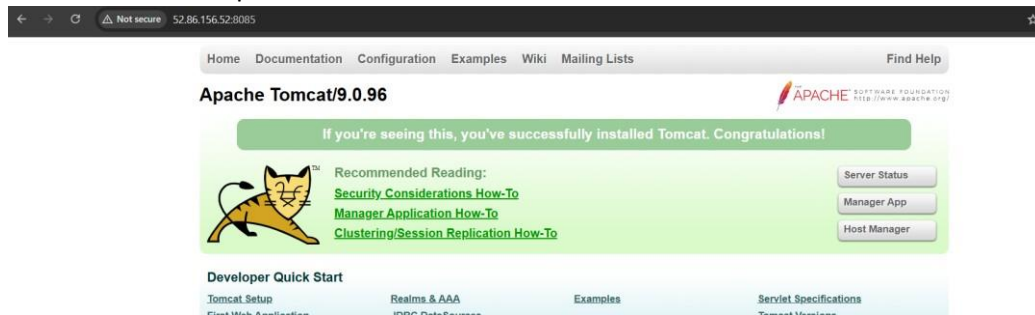
➤ Run docker container and check with ps

```
ubuntu@ip-172-31-25-147:~/alpine-tomcat$ docker run -d \
--name alpine-tomcat-container \
-p 8085:8080 \
alpine-tomcat:v1
c01b14ffcbecaefc34a9592808c277e0b02a45314f6d65e08766939d66be0778
ubuntu@ip-172-31-25-147:~/alpine-tomcat$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
c01b14ffcbec	alpine-tomcat:v1	"/tomcat/bin/catalin..."	15 seconds ago	Up 14 seconds	0.0.0.0:8085->8080

```
ubuntu@ip-172-31-25-147:~/alpine-tomcat$
```

➤ Check with port no 8085



4. Create single stage and multi stage docker file using this source code:

<https://github.com/betawins/multi-stage-example.git>

➤ Check git version and clone Repo and

```
ubuntu@ip-172-31-25-147:~$ git --version
git version 2.53.0
ubuntu@ip-172-31-25-147:~$ git clone https://github.com/betawins/multi-stage-example.git
Cloning into 'multi-stage-example'...
remote: Enumerating objects: 31, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 31 (delta 2), reused 1 (delta 1), pack-reused 24 (from 1)
Receiving objects: 100% (31/31), 53.25 KiB | 8.88 MiB/s, done.
Resolving deltas: 100% (3/3), done.
ubuntu@ip-172-31-25-147:~$ cd multi-stage-example
ubuntu@ip-172-31-25-147:~/multi-stage-example$ ls
Dockerfile  README.md  mvnw  mvnw.cmd  pom.xml  src
```

- Create Single Stage Docker file

```
aws | Search

FROM maven:3.9.6-eclipse-temurin-17

WORKDIR /app

COPY . .

RUN mvn clean package

EXPOSE 8080

CMD ["java", "-jar", "target/demo.jar"]
```

- Now build single stage docker

```
ubuntu@ip-172-31-25-147:~/multi-stage-example$ ls
Dockerfile Dockerfile.single README.md mvnw mvnw.cmd pom.xml src
ubuntu@ip-172-31-25-147:~/multi-stage-example$ vi Dockerfile.single
ubuntu@ip-172-31-25-147:~/multi-stage-example$ docker build -f Dockerfile.single -t single-stage-app:v1 .
```

- Successfully build

```
[INFO] Replacing main artifact with repackaged archive
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 22.296 s
[INFO] Finished at: 2026-06-02T13:34:09Z
[INFO] -----
---> Removed intermediate container 0d6284fc040f
---> 3ec3158faef0
Step 5/6 : EXPOSE 8080
---> Running in 955f11e8eb1a
---> Removed intermediate container 955f11e8eb1a
---> 69974bd10a15
Step 6/6 : CMD ["java", "-jar", "target/demo.jar"]
---> Running in a5d9f07cfd0a
---> Removed intermediate container a5d9f07cfd0a
---> 5558c7d54c71
Successfully built 5558c7d54c71
Successfully tagged single-stage-app:v1
```

- Check with docker images single stage docker exist

```
ubuntu@ip-172-31-25-147:~/multi-stage-example$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
928430096503.dkr.ecr.us-east-1.amazonaws.com/aws.config:latest	371e84608aa7	584MB	195MB	U
alpine-tomcat:v1	f61ad2630e2b	494MB	178MB	U
alpine:latest	5b10f432ef3d	13.1MB	3.95MB	
amazonlinux:latest	ad882807c6c8	227MB	59.3MB	
arunsail7/techie19:v1	925917d788e0	584MB	195MB	
custom-nginx-backup:v1	22764ef71228	584MB	195MB	
custom-nginx:v1	371e84608aa7	584MB	195MB	U
hello-world:latest	0e760fd9fbc48	25.9kB	9.49kB	U
jenkins/jenkins:latest	d566c401e890	801MB	290MB	U
maven:3.9.6-eclipse-temurin-17	29a1658b1f30	728MB	221MB	
nginx:latest	5aca99593157	241MB	66MB	U
single-stage-app:v1	5558c7d54c71	871MB	283MB	
sonarqube:latest	35bedac3f40c	2.52GB	1.08GB	
ubuntu:20.04	5598b87622cd	584MB	150MB	U

- Run container with new port no 8087

```
ubuntu@ip-172-31-25-147:~/multi-stage-example$ docker run -d \
--name single-stage-container \
-p 8087:8080 \
single-stage-app:v1
73784b609c2caa770ed0bdad76f0f04a5ca31b422dfa5ec2d7ce7843728d4f08
ubuntu@ip-172-31-25-147:~/multi-stage-example$
```

- Create multi vi docker file and type the below script

```
ubuntu@ip-172-31-25-147:~/multi-stage-example$ vi Dockerfile.multi
ubuntu@ip-172-31-25-147:~/multi-stage-example$
```

```
! 007cc5d78c5b2b15 (Docker console)
```

```
# Build Stage
FROM maven:3.9.6-eclipse-temurin-17 AS builder

WORKDIR /app

COPY . .

RUN mvn clean package

# Runtime Stage
FROM eclipse-temurin:17-jre

WORKDIR /app

COPY --from=builder /app/target/multi-stage-example-0.0.1-SNAPSHOT.jar demo.jar

EXPOSE 8080

CMD ["java", "-jar", "demo.jar"]
```

- Build it with this command

docker build -f dockerfile.multi -t multi-stage-app:v1 .

```
ubuntu@ip-172-31-25-147:~/multi-stage-example$ vi Dockerfile.multi
ubuntu@ip-172-31-25-147:~/multi-stage-example$ docker build -f Dockerfile.multi -t multi-stage-app:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 211.5kB
Step 1/9 : FROM maven:3.9.6-eclipse-temurin-17 AS builder
--> 29a1658b1f30
Step 2/9 : WORKDIR /app
--> Using cache
--> 6d1996d59b3c
Step 3/9 : COPY . .
--> f8ac31d2798e
Step 4/9 : RUN mvn clean package
--> Running in 1f82bee5a88e
[INFO] Scanning for projects...
```

➤ Build Successfully

```
---> Removed intermediate container 151913b71840
---> bffef121e15c4
Step 9/9 : CMD ["java", "-jar", "demo.jar"]
---> Running in f80efa0ff626
---> Removed intermediate container f80efa0ff626
---> 257ec9a46506
Successfully built 257ec9a46506
Successfully tagged multi-stage-app:v1
ubuntu@ip-172-31-25-147:~/multi-stage-example$
```

➤ Run Container

```
ubuntu@ip-172-31-25-147:~/multi-stage-example$ docker run -d \
--name multi-stage-container \
-p 8088:8080 \
multi-stage-app:v1
b60ebe8016a054a74cf4e94dad5908ef2117d6b5eae9856ce1e4ee210074aed6
ubuntu@ip-172-31-25-147:~/multi-stage-example$
```

➤ Verify Logs

```
ubuntu@ip-172-31-25-147:~/multi-stage-example$ docker logs -f multi-stage-container

:: Spring Boot :: (v2.2.6.RELEASE)

2026-06-02 16:38:12.652 INFO 1 --- [main] c.t.m.MultiStageExampleApplication : Starting MultiStageExampleApplication v0.0.1-SNAPSHOT on b60ebe8
th PID 1 (/app/demo.jar started by root in /app)
2026-06-02 16:38:12.659 INFO 1 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : No active profile set, falling back to default profiles: default
2026-06-02 16:38:14.175 INFO 1 --- [main] o.apache.catalina.core.StandardService : Starting service [Tomcat]
2026-06-02 16:38:14.186 INFO 1 --- [main] org.apache.catalina.core.StandardEngine : Starting Servlet engine: [Apache Tomcat/9.0.33]
2026-06-02 16:38:14.267 INFO 1 --- [main] o.a.c.c.C.[Tomcat].[localhost].[/] : Initializing Spring embedded WebApplicationContext
2026-06-02 16:38:14.267 INFO 1 --- [main] o.s.web.context.ContextLoader : Root WebApplicationContext: initialization completed in 1508 ms
2026-06-02 16:38:14.607 INFO 1 --- [main] o.s.s.concurrent.ThreadPoolTaskExecutor : Initializing ExecutorService 'applicationTaskExecutor'
2026-06-02 16:38:14.822 INFO 1 --- [main] o.s.b.a.e.web.EndpointLinksResolver : Exposing 2 endpoint(s) beneath base path '/actuator'
2026-06-02 16:38:14.883 INFO 1 --- [main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tomcat started on port(s): 8080 (http) with context path ''
2026-06-02 16:38:14.886 INFO 1 --- [main] c.t.m.MultiStageExampleApplication : Started MultiStageExampleApplication in 2.815 seconds (JVM runn
```

➤ Open Browser and checked URL

← → ↺ ⚠ Not secure 3.84.110.228:8088

Whitelabel Error Page

This application has no explicit mapping for /error, so you are seeing this as a fallback.

Tue Jun 02 16:40:41 UTC 2026

There was an unexpected error (type=Not Found, status=404).

No message available

5. Install docker compose and execute sample application.

- Update Packages **apt update**

```
ubuntu@ip-172-31-25-147:~$ sudo su -
root@ip-172-31-25-147:~# apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [136 kB]
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 Packages [194 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [50.8 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [29.6 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64v3 Packages [94.2 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [28.1 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [46.0 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/restricted amd64v3 Packages [201 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security InRelease [136 kB]
```

- Install Docker compose

```
11 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@ip-172-31-25-147:~# apt install docker-compose -y
Note, selecting 'docker-compose-v2' instead of 'docker-compose'
Installing:
  docker-compose-v2

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 11
  Download size: 14.8 MB
  Space needed: 66.6 MB / 9401 MB available
```

- Verify Installation and Create Docker compose project

```
root@ip-172-31-25-147:~# docker compose version
Docker Compose version 2.40.3+ds1-0ubuntu1
root@ip-172-31-25-147:~# mkdir docker-compose-project
root@ip-172-31-25-147:~# cd docker-compose-project
```

- Create docker-compose.yml and written script

```
root@ip-172-31-25-147:~/docker-compose-project# vi docker-compose.yml
root@ip-172-31-25-147:~/docker-compose-project#
```

```
version: '3'

services:

  nginx:
    image: nginx
    container_name: nginx-compose
    ports:
      - "84:80"

  tomcat:
    image: tomcat:9
    container_name: tomcat-compose
    ports:
      - "8089:8080"
```

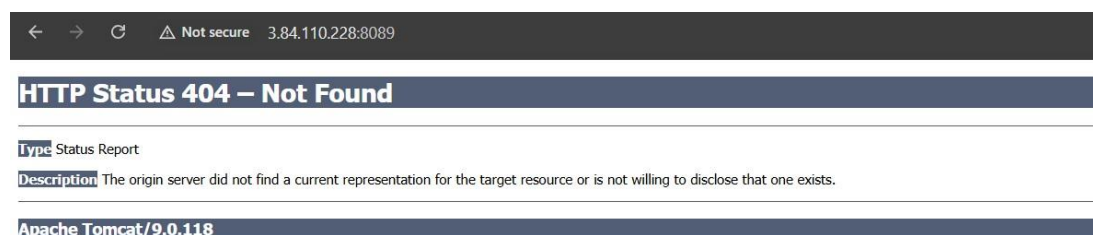
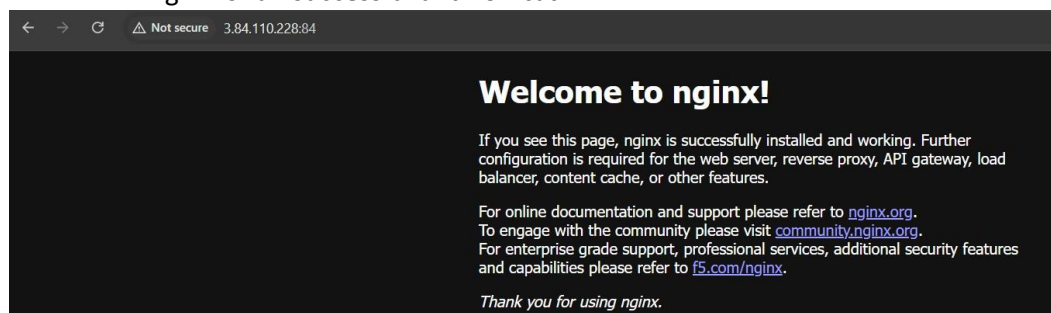
Start Compose

```
root@ip-172-31-25-147:~/docker-compose-project# docker compose up -d
WARN[0000] /root/docker-compose-project/docker-compose.yml: the attribute `version` is obsolete
[+] Running 3/3
 ✓ Network docker-compose-project_default Created
 ✓ Container nginx-compose Started
 ✓ Container tomcat-compose Started
root@ip-172-31-25-147:~/docker-compose-project#
```

➤ Verify Containers

```
root@ip-172-31-25-147:~/docker-compose-project# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
3f888e6d8a5d   nginx         "/docker-entrypoint..." 8 minutes ago  Up 8 minutes  0.0.0.0:84->80/tcp, [:::
7a87769c4035   tomcat:9     "catalina.sh run"         8 minutes ago  Up 8 minutes  0.0.0.0:8089->8080/tcp,
b60ebe8016a0   multi-stage-app:v1 "/_cacert_entrypoint..." 30 minutes ago  Up 30 minutes  0.0.0.0:8088->8080/tcp,
root@ip-172-31-25-147:~/docker-compose-project#
```

➤ Nginx is run successful and Tomcat



6. Implement solution to scan images when pushed to docker registry.

➤ Apt update and Install Trivy and checked version

```
root@ip-172-31-25-147:~# apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 https://aquasecurity.github.io/trivy-repo/deb resolute InRelease
Hit:5 http://security.ubuntu.com/ubuntu resolute-security InRelease
11 packages can be upgraded. Run 'apt list --upgradable' to see them.
root@ip-172-31-25-147:~# apt install trivy -y
trivy is already the newest version (0.71.0).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 11
root@ip-172-31-25-147:~#
```

```
root@ip-172-31-25-147:~# apt install trivy -y
trivy is already the newest version (0.71.0).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 11
root@ip-172-31-25-147:~# trivy --version
Version: 0.71.0
```

➤ Generate Report File

```
root@ip-172-31-25-147:~# trivy image -f table -o report.txt nginx:latest
2026-06-02T17:40:41Z      INFO      [vuln] Vulnerability scanning is enabled
2026-06-02T17:40:41Z      INFO      [secret] Secret scanning is enabled
2026-06-02T17:40:41Z      INFO      [secret] If your scanning is slow, please try '--scanners vuln' to disable secret
2026-06-02T17:40:41Z      INFO      [secret] Please see https://trivy.dev/docs/v0.71/guide/scanner/secret#recommendati
2026-06-02T17:40:41Z      INFO      Detected OS      family="debian" version="13.5"
2026-06-02T17:40:41Z      INFO      [debian] Detecting vulnerabilities...      os_version="13" pkg_num=145
2026-06-02T17:40:41Z      INFO      Number of language-specific files      num=0
2026-06-02T17:40:41Z      WARN      Using severities from other vendors for some vulnerabilities. Read https://trivy.d
ction for details.
root@ip-172-31-25-147:~# ls
docker-compose-project  docker-nginx  report.txt  snap
root@ip-172-31-25-147:~#
```

➤ successfully scanned the Docker image

```
root@ip-172-31-25-147:~# trivy image --severity CRITICAL nginx:latest
2026-06-02T17:44:40Z      INFO      [vuln] Vulnerability scanning is enabled
2026-06-02T17:44:40Z      INFO      [secret] Secret scanning is enabled
2026-06-02T17:44:40Z      INFO      [secret] If your scanning is slow, please try '--scanners vuln' to disable secret scanning
2026-06-02T17:44:40Z      INFO      [secret] Please see https://trivy.dev/docs/v0.71/guide/scanner/secret#recommendation for faster secret detection
2026-06-02T17:44:40Z      INFO      Detected OS      family="debian" version="13.5"
2026-06-02T17:44:40Z      INFO      [debian] Detecting vulnerabilities...      os_version="13" pkg_num=145
2026-06-02T17:44:40Z      INFO      Number of language-specific files      num=0
2026-06-02T17:44:40Z      WARN      Using severities from other vendors for some vulnerabilities. Read https://trivy.dev/docs/v0.71/guide/scanner/vulnerability#seve
ction for details.

Report Summary


| Target                     | Type   | Vulnerabilities | Secrets |
|----------------------------|--------|-----------------|---------|
| nginx:latest (debian 13.5) | debian | 2               | -       |

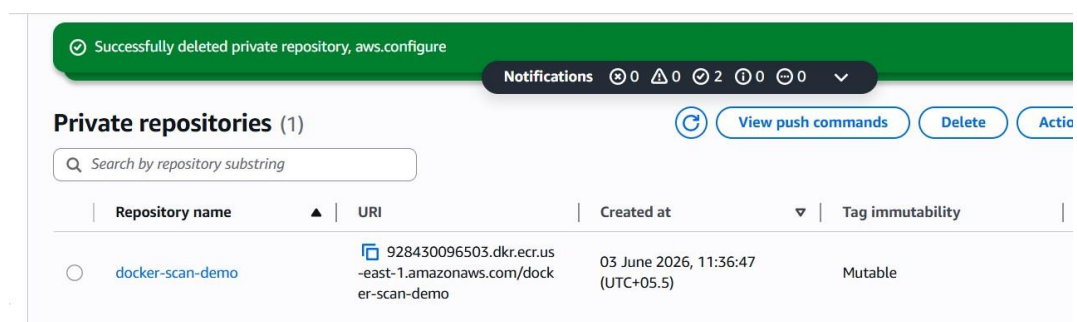

Legend:
- '-': Not scanned
- '0': Clean (no security findings detected)
```

➤ Check Exit Status

```
root@ip-172-31-25-147:~# echo $?
0
root@ip-172-31-25-147:~#
```

7. Implement solution to scan images when pushed to AWS ECR.

➤ Create a new repository in AWS ECR



The screenshot shows the AWS ECR console. At the top, a green notification bar states "Successfully deleted private repository, aws.configure". Below this, the "Private repositories (1)" section is visible. It includes a search bar and a table with the following data:

Repository name	URI	Created at	Tag immutability
docker-scan-demo	928430096503.dkr.ecr.us-east-1.amazonaws.com/docker-scan-demo	03 June 2026, 11:36:47 (UTC+05.5)	Mutable

➤ Login Aws Configure with credentials and login to ECR

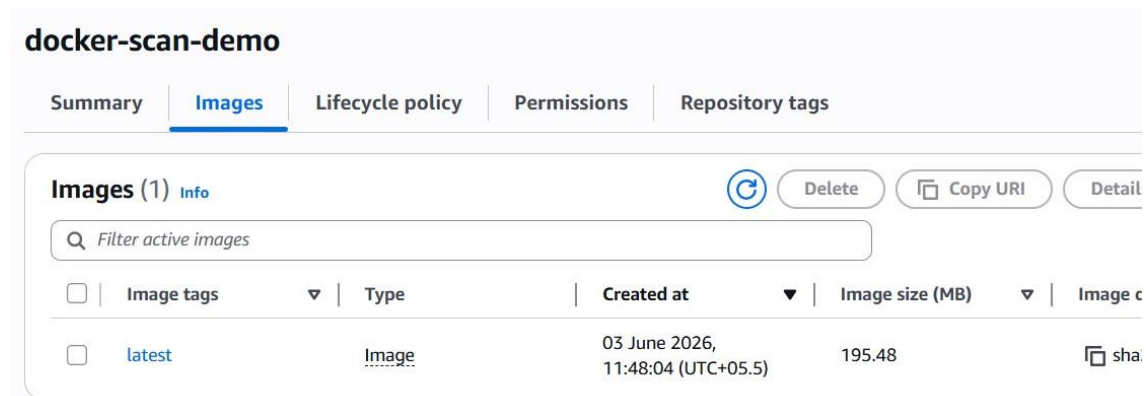
```
ubuntu@ip-172-31-25-147:~$ aws configure
AWS Access Key ID [*****F7W2]: AKIA5QKV7AB3RFRZF7W2
AWS Secret Access Key [*****eh8k]: +aj0q1NJkrI1l1bkVNO6LWqMRW2Sv9TeJB5aAeh8k
Default region name [us-east-1]: us-east-1
Default output format [json]: json
ubuntu@ip-172-31-25-147:~$ aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-s
WARNING! Your credentials are stored unencrypted in '/home/ubuntu/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
ubuntu@ip-172-31-25-147:~$
```

➤ Now Push Image to ECR

```
login succeeded
ubuntu@ip-172-31-25-147:~$ docker tag custom-nginx:v1 928430096503.dkr.ecr.us-east-1.amazonaws.com/docker-scan-demo:latest
ubuntu@ip-172-31-25-147:~$ docker push 928430096503.dkr.ecr.us-east-1.amazonaws.com/docker-scan-demo:latest
The push refers to repository [928430096503.dkr.ecr.us-east-1.amazonaws.com/docker-scan-demo]
93a064f8c941: Pushed
c7345fb8c651: Pushed
bcf200246380: Pushed
latest: digest: sha256:371e84608aa78f1a29a9e310e6f485fff9b5700aaba2c4b8f2497dac153829f8 size: 972
ubuntu@ip-172-31-25-147:~$
```

➤ Checked in Repo



docker-scan-demo

Summary | **Images** | Lifecycle policy | Permissions | Repository tags

Images (1) [Info](#) [Refresh](#) [Delete](#) [Copy URI](#) [Detail](#)

☐	Image tags	Type	Created at	Image size (MB)	Image c
☐	latest	Image	03 June 2026, 11:48:04 (UTC+05.5)	195.48	sha

8. Create a Jenkins pipeline to create a docker image and push the image to Docker hub.

➤ Pull Jenkins and installed it

```
root@ip-172-31-25-147:~# docker pull jenkins/jenkins:lts
lts: Pulling from jenkins/jenkins
3bdbd616c406: Pull complete
cebc1e77ceba: Pull complete
307f8152a55e: Pull complete
5834ae4b8774: Pull complete
6b35bb76b8c8: Pull complete
801814f62c99: Pull complete
c2314d42cca1: Pull complete
76a355a69c2e: Pull complete
9014a3401c36: Pull complete
4c140080734e: Pull complete
dd00b4ealf64: Pull complete
ae218d1cfa6e: Pull complete
653a4e57bccf: Download complete
Digest: sha256:01c992ffef29dcf41c7164e8c16285c657c2368c4943dda8d68c93fdf54447d5
Status: Downloaded newer image for jenkins/jenkins:lts
docker.io/jenkins/jenkins:lts
root@ip-172-31-25-147:~# docker run -d \
--name jenkins \
-p 8080:8080 \
-p 50000:50000 \
-v jenkins_home:/var/jenkins_home \
-v /var/run/docker.sock:/var/run/docker.sock \
jenkins/jenkins:lts
0fcaa0203d459266682dddee245ffa53d654c5fd1df005a4d19c78dbd32f7b6c
```

➤ Checked running containers Jenkins

```
root@ip-172-31-25-147:~# docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED        STATUS        PORTS
0fcaa0203d45   jenkins/jenkins:lts  "/usr/bin/tini -- /u..."  14 seconds ago  Up 13 seconds  0.0.0.0:8080->8080, 50000->50000/tcp
root@ip-172-31-25-147:~# docker exec jenkins cat /var/jenkins_home/secrets/initialAdminPassword
001eb9alcea9475f9a62cb93693b76fa
```


- Update it and install docker again

```
root@0fcaa0203d45:/# apt update
apt install docker.io -y
Get:1 http://deb.debian.org/debian trixie InRelease [140 kB]
Get:2 http://deb.debian.org/debian trixie-updates InRelease [47.3 kB]
Get:3 http://deb.debian.org/debian-security trixie-security InRelease [43.4 kB]
Get:4 http://deb.debian.org/debian trixie/main amd64 Packages [9671 kB]
Get:5 http://deb.debian.org/debian trixie-updates/main amd64 Packages [5412 B]
Get:6 http://deb.debian.org/debian-security trixie-security/main amd64 Packages [195 kB]
Fetched 10.1 MB in 1s (12.5 MB/s)
24 packages can be upgraded. Run 'apt list --upgradable' to see them.
Upgrading:
  libcap2  libsystemd0  libudev1

Installing:
  docker.io
```

- Checked docker version

```
exit
root@ip-172-31-25-147:~# docker exec -it jenkins docker version
Client:
 Version:           26.1.5+dfsg1
 API version:       1.45
 Go version:        go1.24.4
 Git commit:        a72d7cd
 Built:             Sat May  9 11:34:09 2026
 OS/Arch:           linux/amd64
 Context:           default
```

- Connect Jenkins with root and update it restart Jenkins

```
root@ip-172-31-25-147:~# docker exec -u root -it jenkins bash
root@0fcaa0203d45:/# apt update
apt install docker.io -y
Hit:1 http://deb.debian.org/debian trixie InRelease
Hit:2 http://deb.debian.org/debian trixie-updates InRelease
Hit:3 http://deb.debian.org/debian-security trixie-security InRelease
21 packages can be upgraded. Run 'apt list --upgradable' to see them.
docker.io is already the newest version (26.1.5+dfsg1-9+b13).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 21
root@0fcaa0203d45:/# exit
exit
root@ip-172-31-25-147:~# docker restart jenkins
jenkins
root@ip-172-31-25-147:~# docker exec -it jenkins docker version
Client:
 Version:           26.1.5+dfsg1
 API version:       1.45
```


- Give permissions to docker and checked version

```
al/run/docker.sock: connect: permission denied
root@ip-172-31-25-147:~# chmod 666 /var/run/docker.sock
root@ip-172-31-25-147:~# docker exec -it jenkins docker version
Client:
 Version:           26.1.5+dfsg1
 API version:       1.45
 Go version:        go1.24.4
 Git commit:        a72d7cd
 Built:             Sat May  9 11:34:09 2026
 OS/Arch:           linux/amd64
 Context:           default

Server:
 Engine:
  Version:          29.1.3
```

- Created index and docker file and save it

```
cd docker-project
root@ip-172-31-25-147:~/docker-project# vi index.html
root@ip-172-31-25-147:~/docker-project# vi Dockerfile
root@ip-172-31-25-147:~/docker-project# ls
Dockerfile index.html
root@ip-172-31-25-147:~/docker-project# docker build -t test-nginx .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  3.072kB
Step 1/2 : FROM nginx:latest
--> 5aca99593157
Step 2/2 : COPY index.html /usr/share/nginx/html/index.html
--> 1ef27cf52923
Successfully built 1ef27cf52923
Successfully tagged test-nginx:latest
root@ip-172-31-25-147:~/docker-project#
```

- Open Jenkins URL and done with configuration and created a pipeline

 **Jenkins** / All ▾ / New Item

New Item

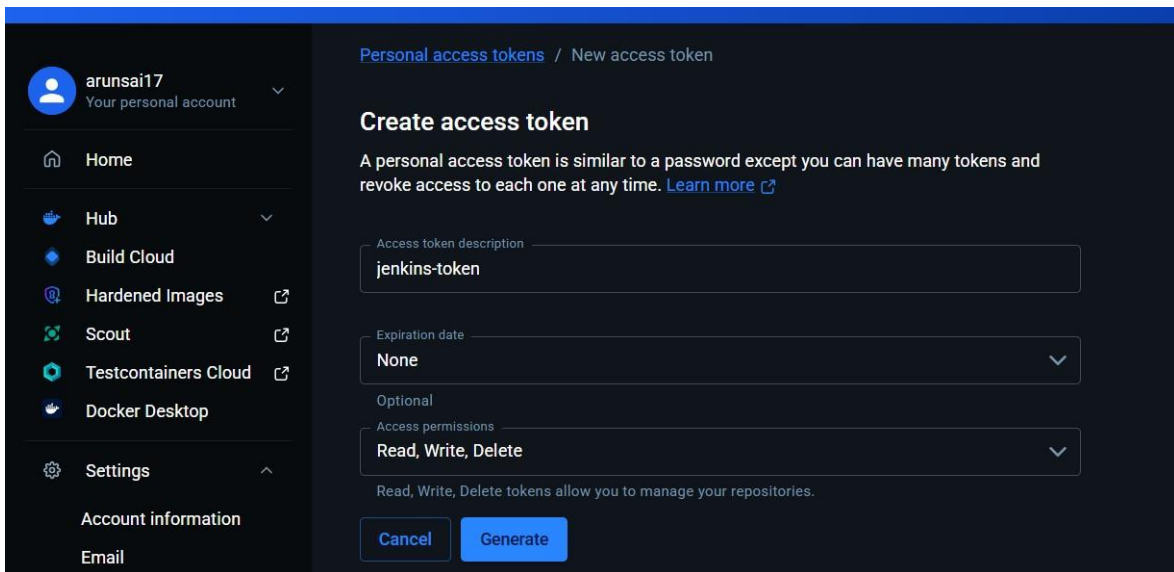
Enter an item name

Select an item type

 Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

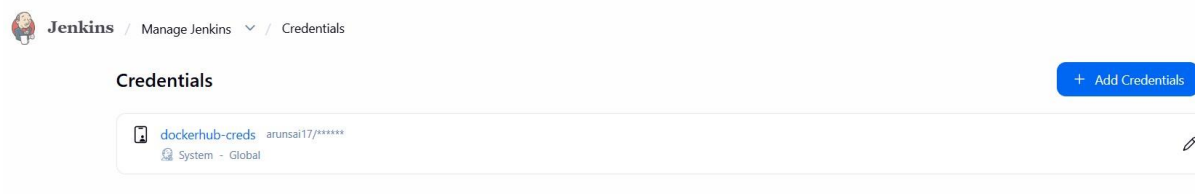
 Freestyle project

➤ In Docker Hub Generate Token



The screenshot shows the Docker Hub interface for a user named 'arunsai17'. The left sidebar contains navigation links: Home, Hub, Build Cloud, Hardened Images, Scout, Testcontainers Cloud, Docker Desktop, Settings, Account information, and Email. The main content area is titled 'Personal access tokens / New access token' and 'Create access token'. It includes a description of personal access tokens, a form to create a new token with fields for 'Access token description' (filled with 'jenkins-token'), 'Expiration date' (set to 'None'), and 'Access permissions' (set to 'Read, Write, Delete'). There are 'Cancel' and 'Generate' buttons at the bottom.

➤ In Jenkins managing open credentials and create new creds for docker



The screenshot shows the Jenkins 'Credentials' page. The breadcrumb trail is 'Jenkins / Manage Jenkins / Credentials'. There is an 'Add Credentials' button. A single credential is listed: 'dockerhub-creds' with a masked value 'arunsai17/*****'. It is categorized as 'System - Global'.

In Pipeline written a pipeline script and save it and build now

```
pipeline {
  agent any

  environment {
    IMAGE_NAME = "arunsai17/custom-nginx:v1"
  }

  stages {

    stage('Create Files') { steps {

      writeFile file: 'Dockerfile', text: '''
      FROM nginx:latest

      COPY index.html /usr/share/nginx/html/index.html
      '''

      writeFile file: 'index.html', text: ''
```

<h1>Jenkins Docker Pipeline Success</h1>

'''

}

}

stage('Build Docker Image') {

steps {

sh 'docker build -t \$IMAGE_NAME .'

}

}

stage('Docker Login') { steps {

withCredentials([usernamePassword(credentialsId:

'dockerhub-creds', usernameVariable: 'DOCKER_USER',

passwordVariable: 'DOCKER_PASS'

))] {

sh 'echo \$DOCKER_PASS | docker login -u \$DOCKER_USER --password-stdin'

}

}

}

stage('Push Docker Image') { steps {

sh 'docker push \$IMAGE_NAME'

}

}

}

post {

success {

echo 'Docker Image Successfully Pushed to Docker Hub'

}

failure {

echo 'Pipeline Failed'

}

}

}

➤ Success Build

 **Jenkins** / docker-pipeline ▾ / #15 ▾

```
830625e1ac85: Mounted from library/nginx
25f84a020409: Waiting
5b4d6ff92fc4: Mounted from library/nginx
45381ecb0e2f: Mounted from library/nginx
13fd728be9eb: Mounted from library/nginx
7f8b1a2b17d8: Mounted from library/nginx
3558722e74b5: Pushed
25f84a020409: Pushed
v1: digest: sha256:9943b2ca5760433d5cf6820a32744f33c56d13f1a4f077e8bc870d55acd6cae4 size: 856
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Docker Image Successfully Pushed to Docker Hub
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

➤ Check in Docker Hub Repo

 Explore My Hub

Search Docker Hub CtrlK    

 **arunsai17**
Your personal account ▾

 Repositories
 Hardened Images
 Collaborations
 Settings
Default privacy
Notifications
 Billing

Repositories

All repositories within the arunsai17 namespace.

All content ▾

Name	Last Pushed ↑	Contains	Visibility
arunsai17/custom-nginx	3 minutes ago	IMAGE	Public
arunsai17/techie19	2 days ago	IMAGE	Public